Introduction & Objectives

1. Background of the Project

E-commerce platforms generate a continuous flow of business data through customer orders, products, sellers, payments, reviews, and delivery operations. This data contains information about customer behaviour, sales performance, product categories, transaction values, delivery timelines, and customer satisfaction.

Analysing such data is important for understanding business performance because raw transactional data alone does not directly provide meaningful business insights. Data analysis can help identify sales trends, customer purchasing behaviour, high-performing products, delivery performance, payment preferences, and areas requiring operational improvement.

The **E-Commerce Business Intelligence and Customer Analytics System** is proposed as a data analytics and business intelligence project that collects, cleans, analyses, and visualizes e-commerce data using Python, SQL, and Power BI. The project focuses on practical application of data analysis, database querying, business intelligence, data visualization, and analytical reporting concepts.

The project uses the Brazilian E-Commerce Public Dataset by Olist, containing information related to customers, orders, order items, products, sellers, payments, reviews, and geographic locations. The system will transform the raw datasets into structured analytical information and interactive dashboards for business decision-making.

2. Problem Statement

Large e-commerce datasets contain thousands of transactions and multiple related tables. Manually examining such data can be difficult because important information is distributed across customers, orders, products, payments, sellers, and review records.

A business user may not easily identify patterns such as changes in sales over time, high-performing product categories, repeat customer behaviour, delayed deliveries, preferred

payment methods, or the relationship between delivery performance and customer satisfaction.

The proposed system addresses this problem by providing a structured analytical solution for data cleaning, business analysis, SQL-based querying, customer analysis, sales analysis, operational analysis, and dashboard-based visualization. The system will convert raw e-commerce data into meaningful business metrics and visual insights that can support better understanding of business performance.

3. Objectives & Goals

- To develop an e-commerce business intelligence and customer analytics system.
- To collect and analyse structured e-commerce transaction data.
- To clean and preprocess raw datasets using Python.
- To analyse sales, orders, products, customers, payments, sellers, and delivery performance.
- To identify important business trends and patterns from historical e-commerce data.
- To use SQL queries for business-oriented data analysis.
- To analyse customer purchasing behaviour and identify repeat customers.
- To evaluate product and category performance.
- To analyse delivery performance and identify delayed orders.
- To study customer review scores and satisfaction patterns.
- To develop interactive dashboards using Microsoft Power BI.
- To present analytical findings through appropriate charts, KPIs, and visualizations.
- To apply data analytics and business intelligence concepts studied during the course in a practical project.

Tools & Technologies

Programming and Data Technologies

Python: Python is used for data loading, cleaning, preprocessing, exploratory data analysis, feature creation, statistical analysis, and analytical processing.

Pandas: Pandas is used for manipulating structured datasets, handling missing values, filtering records, merging tables, grouping data, and creating analytical datasets.

NumPy: NumPy is used for numerical operations and supporting data-analysis calculations.

SQL: SQL is used to query structured e-commerce data and perform business-oriented analysis involving filtering, aggregation, grouping, joins, subqueries, and analytical calculations.

Jupyter Notebook: Jupyter Notebook is used for documenting and executing the Python-based data analysis process in an interactive environment.

Frameworks and Libraries

Matplotlib: Used for creating analytical charts and visual representations of trends and patterns.

Seaborn: Used for statistical visualization and exploratory data analysis.

Power BI: Microsoft Power BI is used to create interactive dashboards, KPI cards, charts, filters, and business intelligence reports.

Software Requirements

Software	**Purpose**
Python 3.x	Data processing and analysis
Pandas	Data manipulation and preprocessing
NumPy	Numerical analysis
Jupyter Notebook	Interactive data analysis
Matplotlib / Seaborn	Data visualization
SQL	Database querying and analysis
Microsoft Power BI	Dashboard and business intelligence
Visual Studio Code	Project development
Git / GitHub	Version control and project publication

Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB or above
Storage	At least 10 GB free space
Display	Standard HD display
Network	Wi-Fi or Ethernet network interface
Input	Standard keyboard and mouse

Problem Definition & Planning

Comparison of Existing Approach and Developed System

Area	Existing Approach	Developed System
Data Analysis	Raw datasets may require manual examination.	Provides structured analysis using Python and SQL.
Data Cleaning	Missing or inconsistent data may be difficult to identify manually.	Performs systematic data-quality assessment and preprocessing.
Sales Analysis	Sales information may be distributed across multiple datasets.	Combines relevant tables to analyse sales performance.
Customer Analysis	Customer behaviour may be difficult to identify from raw transactions.	Analyses purchasing behaviour and repeat purchases.
Product Analysis	Product performance may require manual calculations.	Provides category and product-level performance analysis.
Delivery Analysis	Delivery performance may be difficult to evaluate.	Calculates delivery time and delivery-delay metrics.
Customer Satisfaction	Reviews may exist separately from order information.	Connects review information with order and delivery analysis.
Visualization	Information may remain in tables or spreadsheets.	Presents KPIs and trends through interactive Power BI dashboards.
Business Decision Support	Requires manual interpretation of raw information.	Provides structured analytical insights for decision-making.

Functional Requirements

Function	Description
Data Loading	Loads e-commerce datasets into the analytical environment.
Data Profiling	Examines dataset structure, columns, data types, missing values and duplicates.

Function	Description
Data Cleaning	Handles missing, inconsistent and unsuitable data according to analytical requirements.
Data Transformation	Converts and prepares raw data for analysis.
Data Integration	Combines relevant customer, order, product, payment, seller and review information.
Sales Analysis	Analyses revenue, orders, products and sales trends.
Customer Analysis	Analyses customer purchasing behaviour and repeat purchases.
Product Analysis	Identifies high-performing products and categories.
Delivery Analysis	Calculates delivery duration, delivery delays and delivery performance.
Review Analysis	Analyses customer review scores and satisfaction patterns.
SQL Analysis	Performs business-oriented analysis using SQL queries.
Dashboard	Displays business KPIs, trends, charts and analytical findings.
Reporting	Provides structured analytical results for business interpretation.

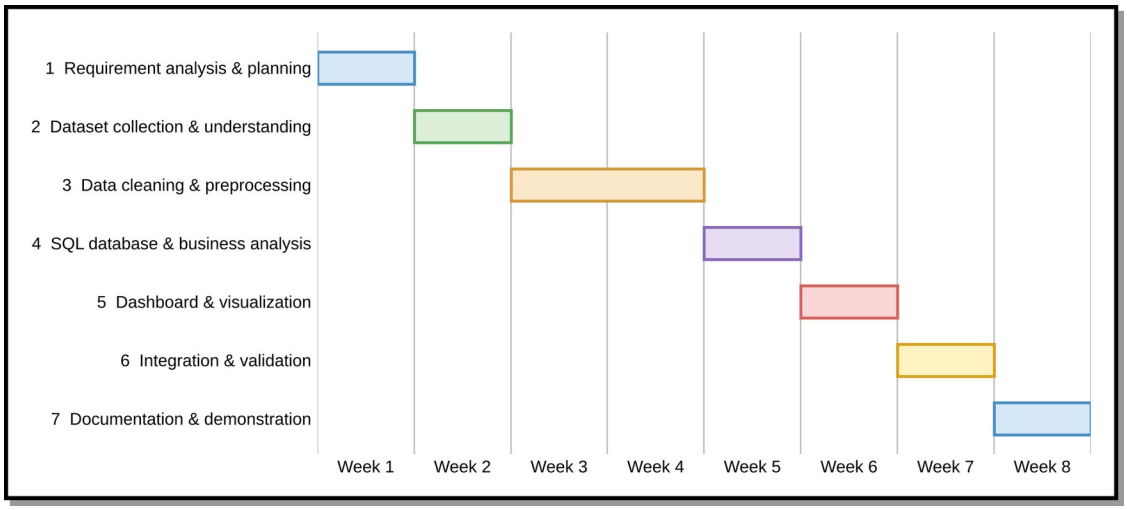
Technical Specifications

- The system will use Python as the primary programming and analytical language.
- Pandas and NumPy will be used for data manipulation and numerical analysis.
- Jupyter Notebook will be used for exploratory data analysis and documentation.
- SQL will be used for structured business queries and relational analysis.
- Related datasets will be joined using appropriate keys such as customer ID, order ID, product ID, and seller ID.
- Data preprocessing will include data-type conversion, missing-value assessment, duplicate checking, and consistency validation.
- Derived analytical fields will be created where required, such as delivery duration and delivery delay.
- Power BI will be used for interactive business dashboards.

- The dashboard will display business KPIs, trends, category performance, customer analysis, and operational metrics.
- The project will be tested by validating calculated metrics and comparing analytical outputs across Python, SQL, and Power BI.
- The project will use the Olist Brazilian E-Commerce Public Dataset for academic and analytical purposes.

Project Timeline

Phase	Activity	Duration
1	Requirement analysis and project planning	Week 1
2	Dataset collection and data understanding	Week 2
3	Data cleaning and preprocessing	Weeks 3–4
4	SQL database and business-analysis development	Week 5
5	Dashboard and data-visualization development	Week 6
6	System integration and analytical validation	Week 7
7	Documentation, refinement and final demonstration	Week 8



Project Timeline (Gantt Chart)

System Analysis

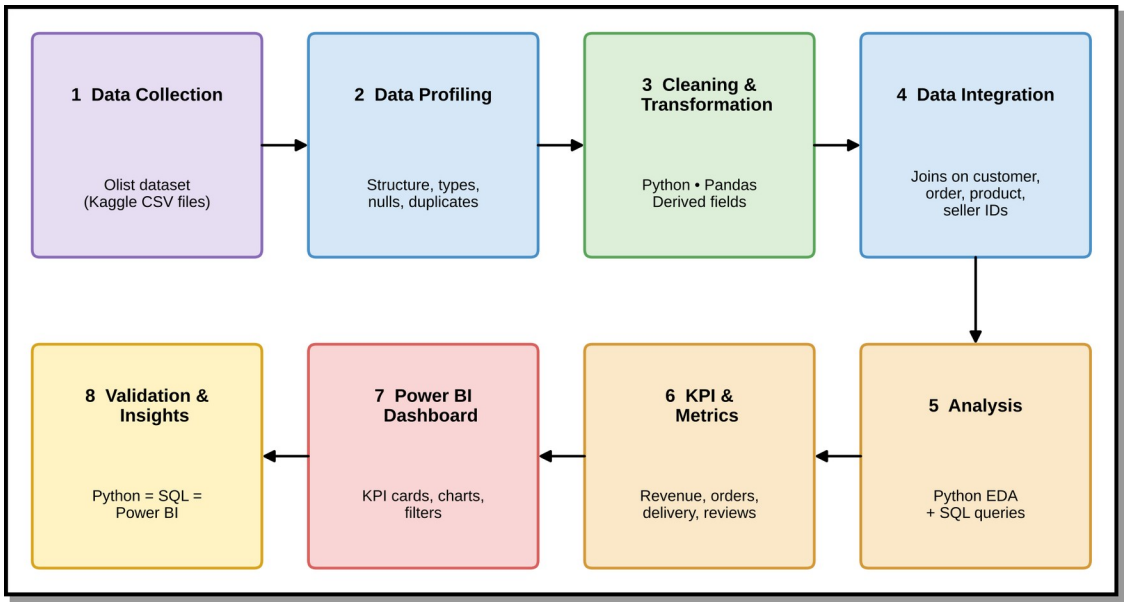
System Overview

The proposed E-Commerce Business Intelligence and Customer Analytics System follows a structured data-analysis architecture. The raw e-commerce datasets are first

collected and loaded into the analytical environment. The data is then profiled to understand its structure, data types, missing values, duplicate records, and relationships between different datasets.

The cleaned and transformed data is analysed using Python and SQL. Relevant customer, order, product, payment, seller, delivery, and review information is combined to generate business metrics and analytical findings.

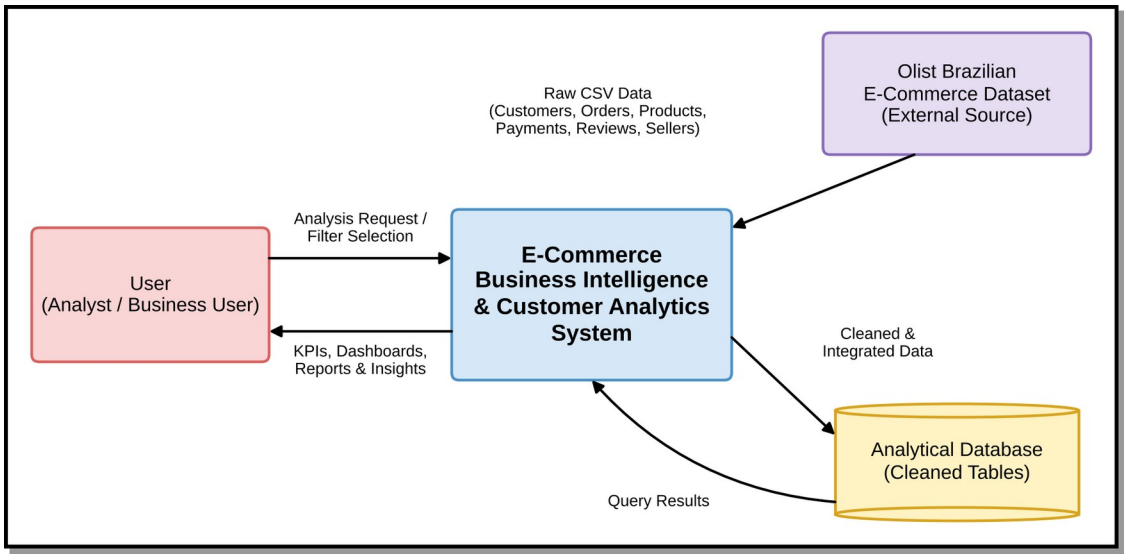
The resulting information is presented through Power BI dashboards containing key performance indicators, charts, trends, filters, and business insights. The overall system therefore combines data collection, preprocessing, analysis, SQL querying, business intelligence, and visualization into one analytical workflow.



Analytical Workflow of the System

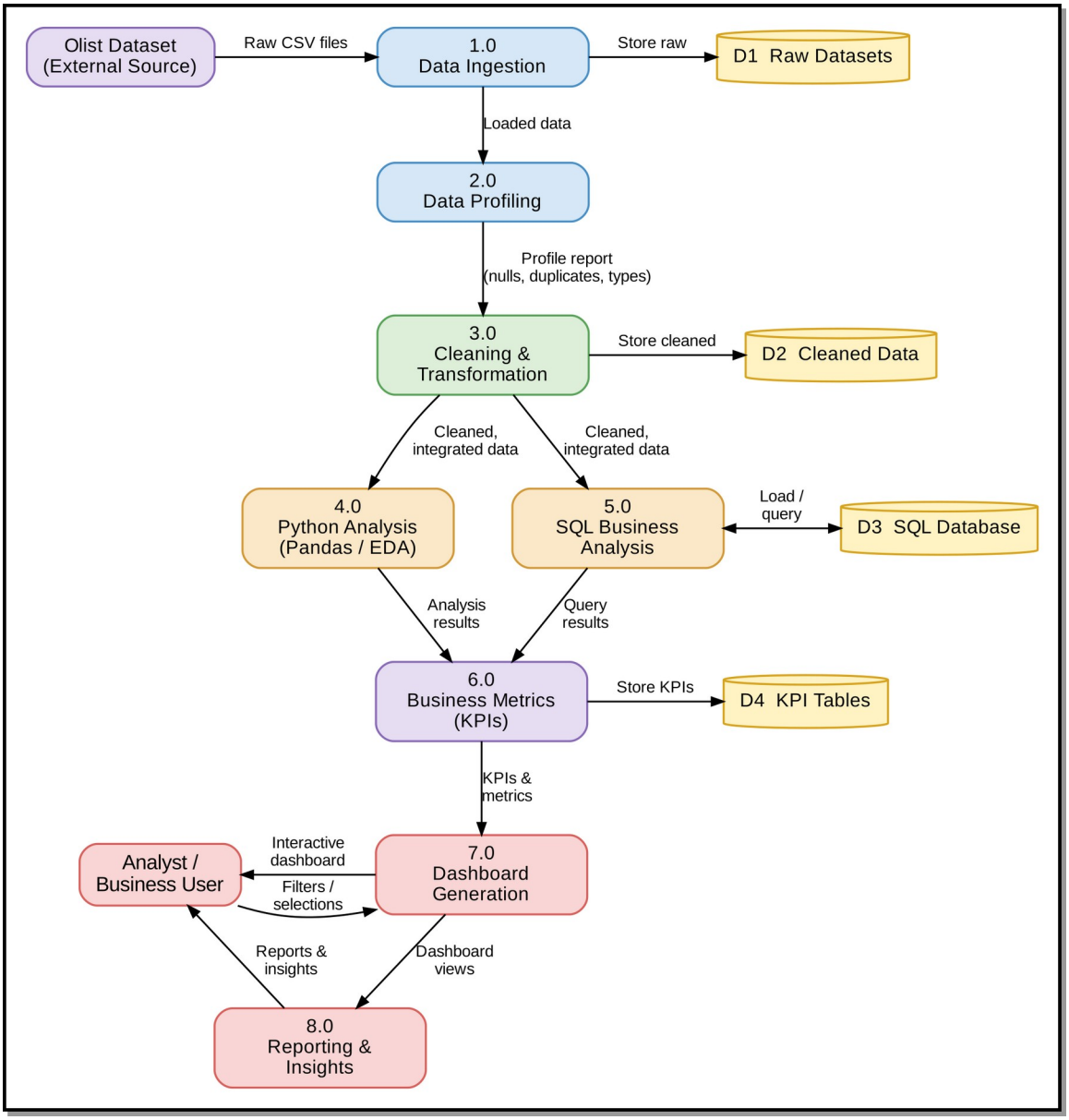
Data Flow Diagram

Level 0: The system receives e-commerce datasets, processes and analyses them, and provides analytical results and dashboards to the analyst/business user.



Level 0 Data Flow Diagram

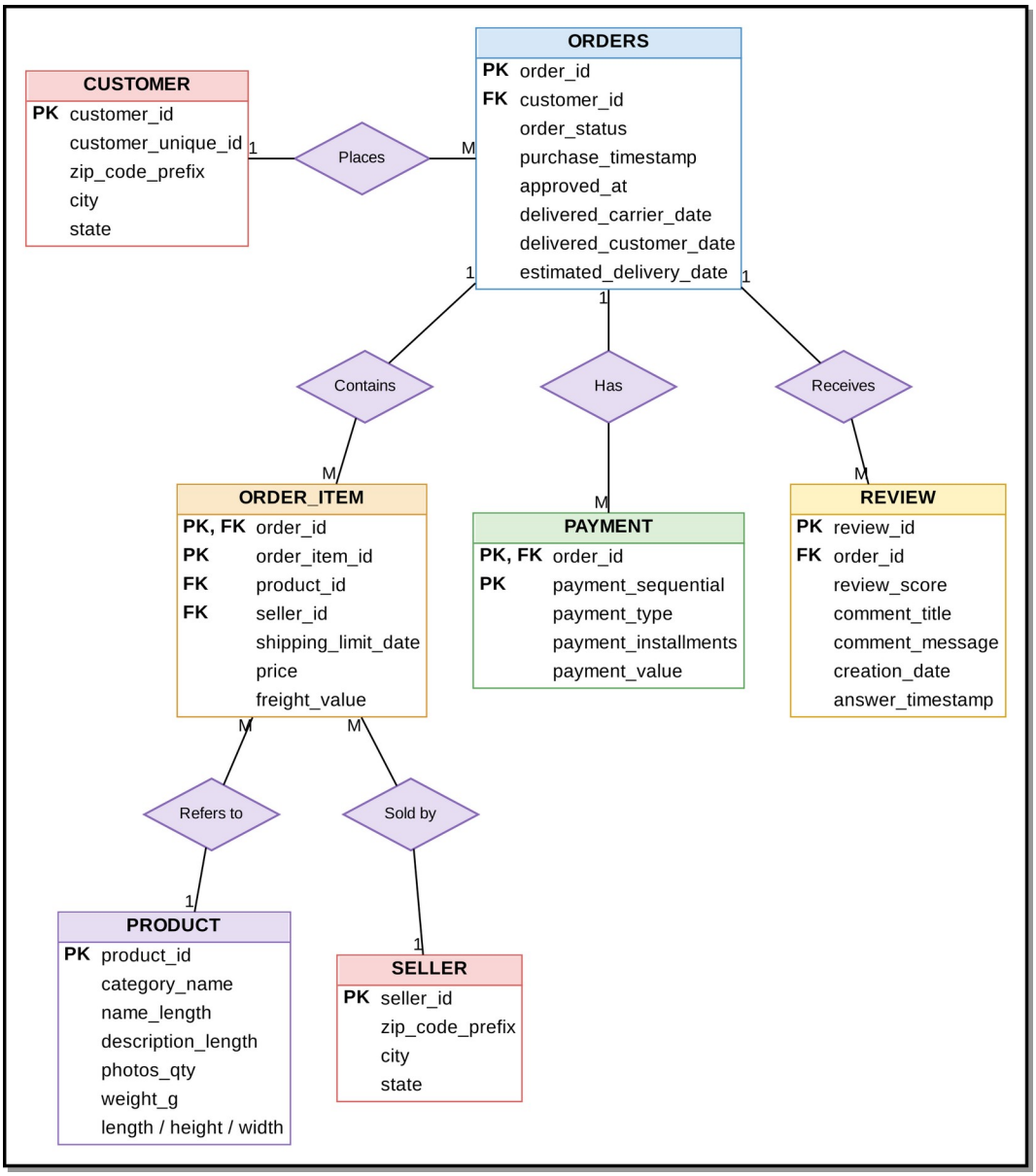
Level 1: The system is decomposed into data ingestion, data profiling, data cleaning and transformation, Python analysis, SQL analysis, business metrics, dashboard generation, and reporting.



Level 1 Data Flow Diagram

Entity Relationship Diagram

The database design represents relationships between customers, orders, order items, products, sellers, payments, and reviews. A customer can be associated with multiple orders. An order can contain multiple order items. Each order item is associated with a product and seller. An order can also have payment and review information.

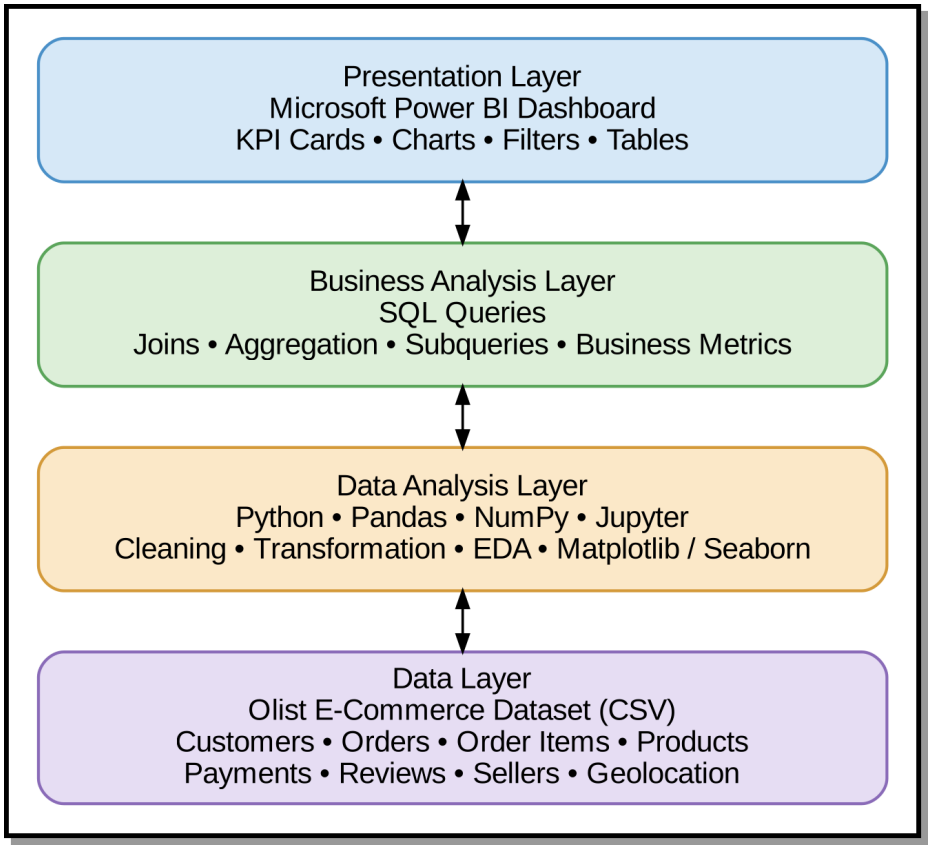


Entity Relationship Diagram

System Design

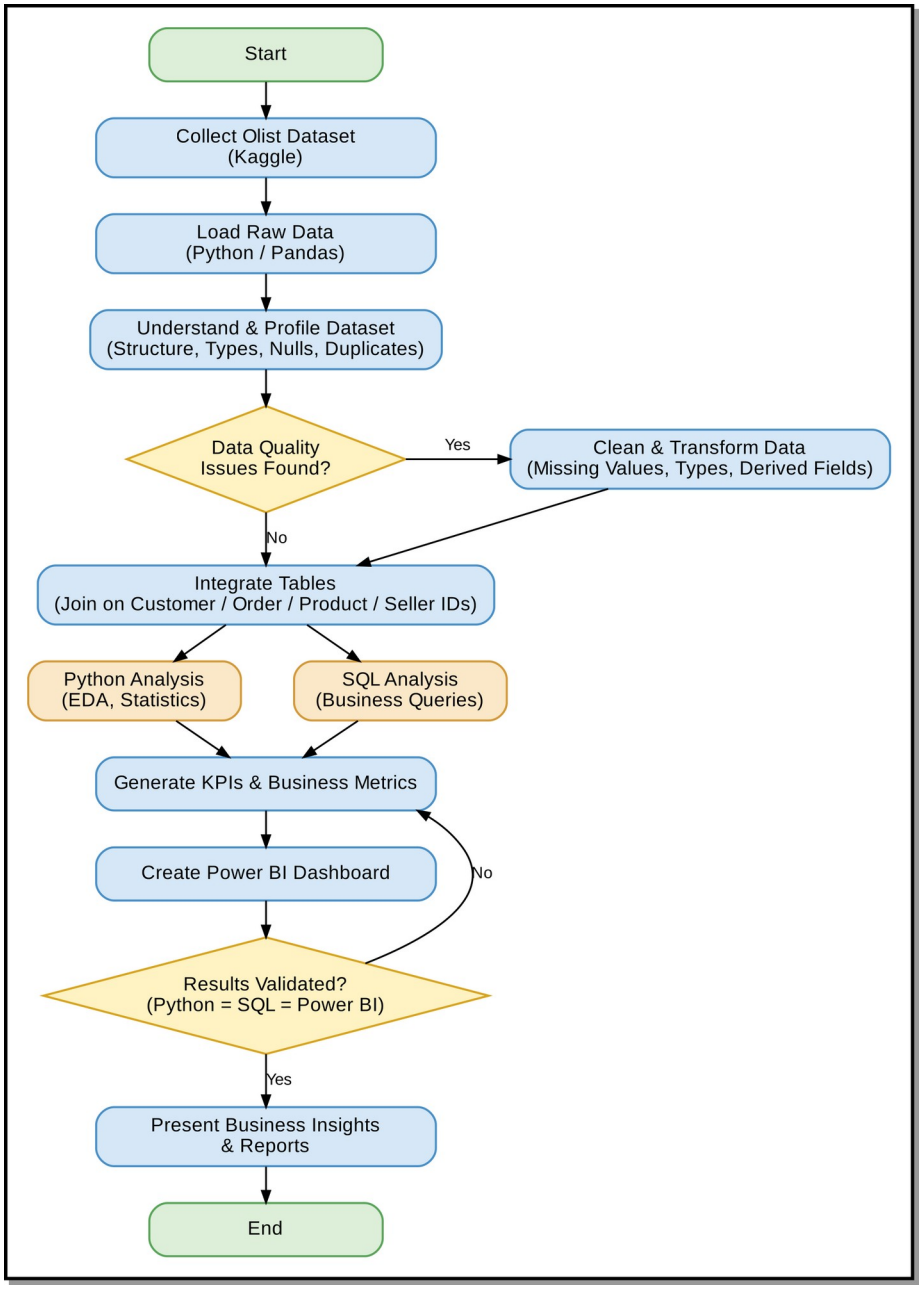
The proposed system is organized into four logical layers:

- Presentation Layer: Provides interactive Power BI dashboards containing KPI cards, charts, filters, tables, and business-performance visualizations.
- Business Analysis Layer: Uses SQL for structured business queries, aggregation, joins, filtering, customer analysis, product analysis, and other analytical calculations.
- Data Analysis Layer: Uses Python for data loading, cleaning, transformation, exploratory data analysis, statistical calculations, and analytical processing.
- Data Layer: Contains structured e-commerce datasets involving customers, orders, products, sellers, payments, reviews, and related information.



Basic System Architecture

The flowchart below shows the step-by-step execution of the system, including the data-quality decision and the final validation check that compares results across Python, SQL, and Power BI.



Basic System Flowchart

Project Modules

The E-Commerce Business Intelligence and Customer Analytics System is divided into modules so that each part performs a specific function and can be developed and tested independently.

Data Collection and Loading Module

Handles collection and loading of the Olist e-commerce datasets containing customer, order, product, seller, payment, review, and geographic information.

Data Quality and Preprocessing Module

Examines missing values, duplicate records, incorrect data types, inconsistent values, potential outliers, and invalid or unexpected records before analysis.

Data Integration Module

Combines related datasets using appropriate keys such as customer identifiers, order IDs, product IDs, and seller IDs.

Sales Analysis Module

Analyses order volume, revenue, sales trends, monthly performance, product category performance, and average order value.

Customer Analytics Module

Analyses customer purchasing behaviour, repeat purchases, purchase frequency, spending behaviour, geographic distribution, and customer segments.

Product and Category Analysis Module

Evaluates product and category performance, including top-selling categories, revenue contribution, product demand and average product price.

Delivery and Operations Analysis Module

Evaluates delivery duration, on-time delivery rate, late delivery rate, delivery delays, and delivery performance by relevant dimensions.

Payment Analysis Module

Analyses payment methods, payment value, installment usage, payment distribution, and payment preferences.

Customer Review Analysis Module

Analyses review scores and explores review performance by product category and delivery performance.

SQL Business Analysis Module

Uses SQL to answer business questions involving sales, customers, products, sellers, geography, and delivery.

Business Intelligence Dashboard Module

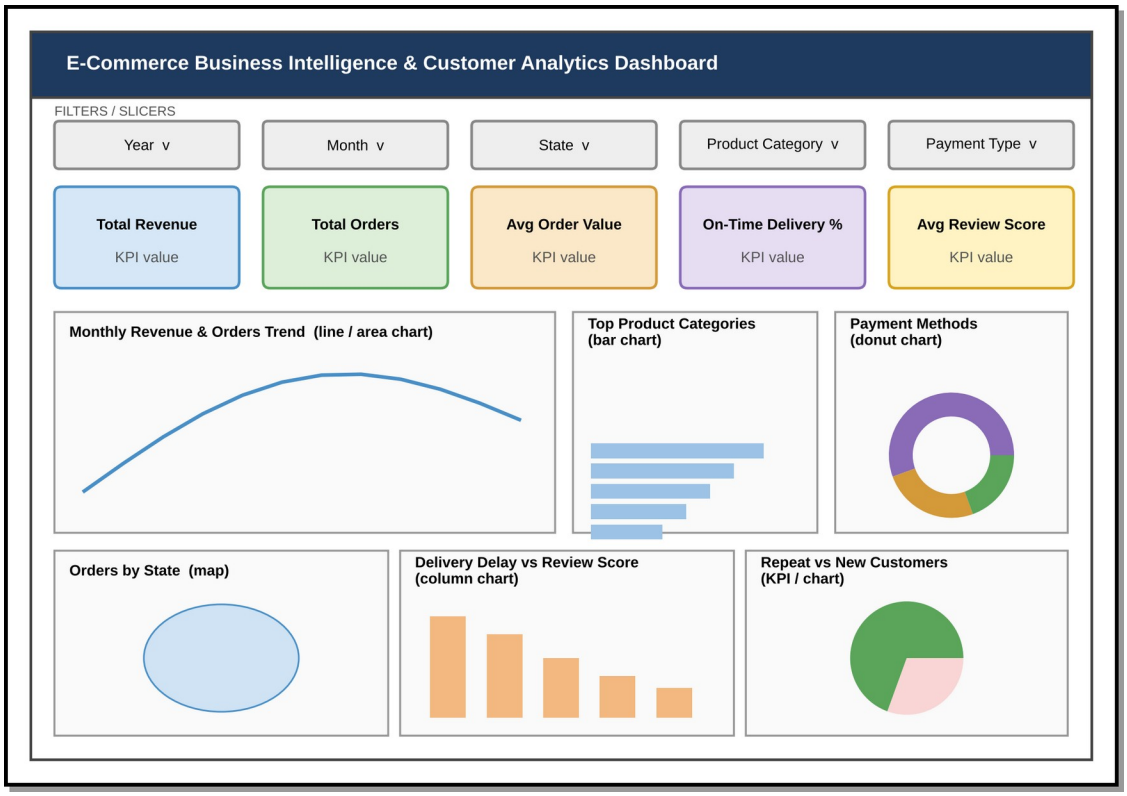
Provides a centralized Power BI view containing KPI cards, trends, category performance, customer analysis, payment analysis, and delivery performance.

Reports and Business Insights Module

Presents final analytical findings in a structured form to identify business trends, operational issues, customer behaviour, and potential areas for improvement.

Proposed Dashboard Layout

The Power BI dashboard is planned as a single-page view with slicers at the top, KPI cards below them, and supporting charts for sales trends, category performance, payment methods, geography, delivery performance and customer behaviour. The layout below is illustrative and will be populated with values from the analysed data.



Proposed Power BI Dashboard Layout

Data Validation and Reliability

Data Validation

- Checking missing values.
- Checking duplicate records.
- Validating data types.
- Checking unique identifiers.
- Validating relationships between tables.
- Comparing Python and SQL results.

- Checking Power BI calculations against source data.
- Reviewing calculated KPIs.

Data Consistency

Related tables will be checked to ensure that identifiers such as order IDs, customer IDs, product IDs, and seller IDs are used consistently. The project will also validate derived metrics such as delivery duration and delivery delay.

Analytical Accuracy

Business metrics displayed in Power BI will be compared with independently calculated Python or SQL results to reduce the possibility of incorrect dashboard calculations.

Controlled Data Usage

The project will use the publicly available Olist Brazilian E-Commerce dataset for academic data-analysis and business-intelligence purposes.

Future Scope

- Machine-learning-based customer segmentation.
- Predictive sales forecasting.
- Customer churn prediction.
- Advanced recommendation-system analysis.
- Automated business reporting.
- Real-time e-commerce analytics.
- Advanced RFM-based customer segmentation.
- Sentiment analysis of customer reviews.
- Predictive delivery-delay analysis.
- Integration with additional e-commerce datasets.
- Cloud-based deployment of the analytics platform.
- Automated refresh of business intelligence dashboards.
- Integration with enterprise business intelligence systems.

- Advanced executive-level reporting and KPI monitoring.

Conclusion

The E-Commerce Business Intelligence and Customer Analytics System is proposed as a practical data analytics and business intelligence project for analysing e-commerce business performance.

The system combines data collection, data preprocessing, Python-based exploratory analysis, SQL-based business analysis, customer analytics, product analysis, delivery analysis, review analysis, and Power BI-based visualization into a single analytical workflow.

The project provides a practical implementation of concepts related to data cleaning, data transformation, exploratory data analysis, relational databases, SQL querying, business intelligence, data visualization, customer analytics, and business decision-making.

The initial implementation is intentionally focused on a manageable set of business and operational analytics so that the complete project can be developed, tested, demonstrated, and understood. Further enhancements such as machine-learning-based forecasting, customer segmentation, sentiment analysis, predictive analytics, automated reporting, and cloud deployment can be considered in future versions.

The project therefore provides practical exposure to the complete workflow followed in modern data analytics projects, from raw data collection and preprocessing to business insights and interactive dashboard development.

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